MANet: Multi-Scale Aware-Relation Network for Semantic Segmentation in Aerial Scenes

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***Abstract*— Semantic segmentation is an important yet unsolved problem in aerial scenes understanding. One of the major challenges is the intense variations of scenes and object scales. In this article, we propose a novel multi-scale aware-relation network (MANet) to tackle this problem in remote sensing. Inspired by the process of human perception of multi-scale (MS) information, we explore discriminative and diverse MS representations. For discriminative MS representations, we propose an inter-class and intra-class region refinement (IIRR) method to reduce feature redundancy caused by fusion. IIRR utilizes the refinement maps with intra-class and inter-class scale variation to guide MS fine-grained features. Then, we propose multi-scale collaborative learning (MCL) to enhance the diversity of MS feature representations. The MCL constrains the diversity of MS feature network parameters to obtain diverse information. Also, the segmentation results are rectified according to the dispersion of the multilevel network predictions. In this way, MANet can learn MS features by collaboratively exploiting the correlation among different scales. Extensive experiments on image and video datasets, which have large-scale variations, have demonstrated the effectiveness of our proposed MANet.**

***Index Terms*— Aware relation, collaborative learning, feature refinement, multi-scale (MS), remote sensing, semantic segmentation.**

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I. Introduction

SEMANTIC segmentation aims to assign semantic categories to pixels in an image [1]. In remote sensing, semantic segmentation in aerial scenes plays an increasingly important role in a wide range of applications, such as land cover classification [2], remote sensing scene understanding [3], and urban planning [4]. Due to the success of deep learning and the astonishing results on several semantic segmentation benchmarks of natural images [5]–[7], most methods for remote sensing image semantic segmentation are based on deep learning architectures [8]–[13]. However, the size of remote sensing images is usually much larger than the typical natural images used in computer vision applications [14], and a wide range of perspectives causes image objects of intense variation scales. In addition, the multi-scale (MS) variation problem is more serious in datasets captured by an unmanned aerial vehicle (UAV), because the oblique view makes objects scale variations at different distances [15], [16].

Many methods explore MS information to address the scale issues. They use varying scale parameters to construct an MS representations in the feature space [17], [18], e.g., image pyramids [19], multigrid [20], [21], and wavelets [22], [23]. Image pyramids methods adjust the sizes of training images with multiple ratios and use parallel multibranch to make the network perceive MS information [24]–[28]. Then, they receive the final segmentation results by fusing MS predictions. Image pyramids are also used for data enhancement for semantic segmentation tasks [1], [29]. Multigrid methods [5], [30]–[32] employ a hierarchy grid of different sizes [33] to capture MS context. In particular, ZigZagNet [30] iteratively fuses intermediate feature maps between the top-down and the bottom-up pathway to aggregate the MS information over multiple stages. Features of skip net are MS, because the size of the receptive field gradually increases with the depth of the network. DeepLabV3+ [5] uses atrous spatial pyramid pooling (ASPP) [34] to exploit MS context information with different receptive fields.

However, these methods are limited to exploring discriminative and diverse MS feature representations. Humans' visual systems explore a wide variety of information at MS and sort out the discriminative information of each scale for reasoning and judgment. Just like this, MS representations are expected to be discriminative and diverse. As illustrated in Fig. 1(a), previous methods usually utilize average pooling to fuse different scaled features. After the fusion, there are a large number

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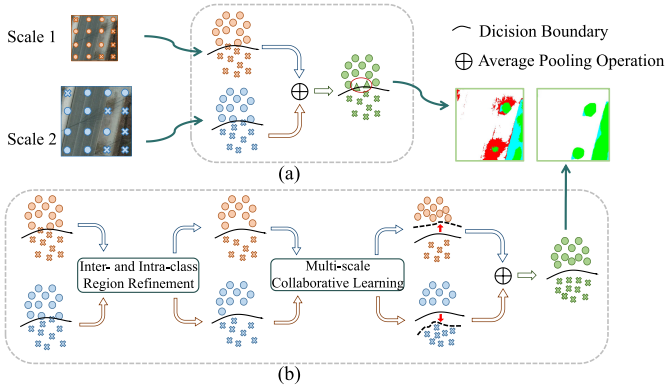


Fig. 1. Comparison of previous and the proposed methods (different shapes represent different classes). (a) Existing methods fuse MS features but can fail by generating ambiguous features near the decision boundary. (b) IIRR reduces the confusing samples near the decision boundary and enhances the discriminant ability of MS feature representations. Then, MCL rectifies MS features from different views (diverse decision boundaries) to achieve compatible fusion.

of confusing representations near the discriminant boundary, which affects the accuracy of semantic segmentation. Some methods [26], [27] use soft weight maps to replace average pooling. The MS method [27] learns relative weights between adjacent scales. The work [35] designs attention modules to refine MS features. The method [36] introduces additional modules to enhance feature diversity. Nevertheless, it is still challenging to represent and fuse MS features effectively.

In this article, we propose to explore discriminative and diverse MS features, as Fig. 1(b) demonstrated. First, inter-class and intra-class region refinement (IIRR) is proposed to enhance the discriminability of MS representations. Concretely, the refinement strategy separately considers the inter-class and intra-class scale variation and utilizes regional high-level semantic representations to refine MS predictions. IIRR reduces the redundancy of fused features and improves the accuracy of semantic segmentation. In addition, to enhance the diversity of MS features, we design the discrepancy classifier for different scale features and minimize the similarity of the discrepancy classifier parameters. Finally, we minimize the uncertainty regions and adaptively rectify the segmentation results at the pixel level.

The main contributions of this article are as follows.

1) A novel multi-scale aware-relation network (MANet) is proposed to learn relations between different scales and explore discriminative and diverse MS representations. IIRR is proposed to reduce feature redundancy and enhance the discriminability of MS representations.
2) Multi-scale collaborative learning (MCL) is proposed to enhance the diversity of MS feature representations. The discrepancy classifiers are designed to enhance the diversity of MS feature representations for semantic segmentation. We minimize uncertainty regions and adaptively rectify the segmentation results at the pixel level.
3) Experiments show the effectiveness of MANet on both image and video datasets that have large-scale variations problems: 1) the Potsdam dataset, which is a widely used benchmark dataset; 2) the National Artificial Intelligence Challenge (NAIC) dataset, which is a multiresolution image dataset for land-used semantic segmentation; and 3) the UAVid dataset, which is a high-resolution UAV video dataset for semantic segmentation tasks focusing on urban scenes.

The remainder of this article is organized as follows. Section II introduces related work on image segmentation using deep learning. Section III describes our MANet in detail. Section IV presents the data and our detailed experimental evaluation. Section V discusses the experimental results. Finally, the conclusions of this article are drawn in Section VI.

II. Related Work

In this section, first, we review the methods of semantic segmentation for natural images; then, we briefly discuss the semantic segmentation of remote sensing data.

A. MS Semantic Segmentation

1) MS Architecture: MS representation has been proven to be effective for many semantic segmentation tasks [31], [32], [37], [38]. There are two main architectures proposed to use MS representations, namely, image pyramids methods and multigrid methods. As for the image pyramids methods, the input images are resized to different scales and use parallel multibranch to capture MS representations. It then fuses the MS predictions with average-pooling or max-pooling operation [24]–[28]. For the multigrid methods, MS features are combined from the intermediate layers of models. Features from different layers, which have different receptive field sizes, are MS in nature. Pyramid scene parsing network (PSPNet) [31] introduces a spatial pyramid pooling module, which captures MS context information by assembling features at four scales using the features obtained from the last layer. DeepLab [34] utilizes exponentially expanded receptive fields to exploit MS context information. More recently, ZigZagNet [30] iteratively fuses intermediate feature maps between the top-down and the bottom-up pathway, not just the features of the last layer to aggregate the MS information over multiple stages. Densely Connected Neural Architecture Search (DCNAS) [39] searches for an optimal MS network structure using neural architecture search (NAS) methods. In DCNAS, a densely connected search space is introduced with an agentless search paradigm.

The proposed network differs from previously proposed related works. Specifically, the MS method [27] learns relative weights between adjacent scales. Difference from this method, the proposed method learns absolute weights at different scales and constrains the feature diversity among different scales. The work [35] designs attention modules to refine high-level semantic features. The difference is that IIRR considers fine-grained details information to help for land objects recognition. Specifically, IIRR refines MS fine-grained features through high-level semantic features, which enhances feature discriminability and computational efficiency. The method [36] proposes an image cascade and feature pyramid network while introducing additional modules to enhance feature diversity.

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The difference is that the proposed method designs a loss function to constrain the diversity of features. The advantage is that the gradient can optimize the entire model parameters through backpropagation, and multiple auxiliary modules in different locations need to be carefully designed.

2) Tri-Training Strategy: Tri-training is proposed in [40] that uses three classifiers to exploit unlabeled data. Tri-training improves co-training [41] that trains alternately to maximize the mutual agreement on distinct views. An algorithm that uses the co-training setting may learn separate classifiers over each feature set and combine the predictions to decrease the error. We do not follow the strategy at the global level. Instead, pixel-level collaborative learning is enforced through co-training. We identify the part whose results are misclassified with large uncertainty, and then encourage the model to focus on this area. To the best of our knowledge, we are making an early attempt to utilize co-training to describe each object at different scale views and measure the discrepancy of MS predictions for diversiform MS representations.

B. Remote Sensing Semantic Segmentation

Deep learning methods have demonstrated much superior performance for semantic segmentation [29], [35], [42], in which it automatically extracts MS features that are tailored for the data [43]–[45]. The great success of deep learning methods excited the development of remote sensing [8], [9].

1) Semantic Segmentation of Remote Sensing Images: Multi-scale convolutional neural network (MCNN) [46] is proposed to learn spatial-related features for hyperspectral remote imagery. MCNN first uses a pyramid structure to present spatial information at multiple scales, and then uses a voting scheme to automatically extract high-level spatial features from MS training dataset. An end-to-end self-cascaded network (scasNet) is proposed in [28], which uses dilated convolutions [34] to capture MS contexts. Adaptive fusion network (AFNet) is proposed in [47] and designs the scale-feature and scale-layer attention module for learning MS features. A two-stage MS architecture is proposed in [14] to learn context information from a global level and learn low-level information from cropped image patches. In this article, we pay attention to exploring discriminative and diverse MS features for semantic segmentation.

2) Semantic Segmentation of Remote Sensing Videos: The use of time information is of great value to predict segmentation results, as the sequence data have consistent predictions across multiple frames. Different frames are provided from different viewing positions, and several predictive signals can be collected by these frames [48]. However, this is also a challenge to semantic video segmentation, because the scales of objects are various in both inter-class and intra-class. To reduce the scale variation effect in datasets captured by an UAV, an MS-dilation net (MSD) is proposed in [15]. But, MSD uses a feature space optimization (FSO) [49] method to smooth the final predictions, which is not an end-to-end approach. In this article, scale variation problems are considered to further improve the semantic segmentation of remote sensing.

III. MS Aware-Relation Network

MS information is known to be significant for semantic segmentation. Merging MS features is an effective operation to aggregate contextual information.

In this section, we present the proposed MANet devised to learn relations between different scales and improve semantic segmentation. The proposed architecture is shown in Fig. 2, which contains three parts. In the first part, a multilevel architecture with different scales is designed to extract MS features. In the second part, IIRR is proposed to make MS features more discriminative. In the third part, MCL is proposed to enhance the diversity of MS feature representations. As a result, MANet can learn MS features simultaneously in a collaborative way that exploits the correlation among different scales.

A. MS Feature Extraction

Given a set of images $X = \{x(i)\}_{i=1}^{N}$ with pixel-level labels $Y = \{y(i)\}_{i=1}^{N}$, the module is expected to extract MS features simultaneously. These simultaneously learned MS features allow us to train the network in a collaborative way.

For each iteration, input image x is resized to different scales $s \in \{1, \ldots, S\}$, denoted as x_s, where S is the total number of scaling levels. Here, we scale down and scale up the original image by bilinear interpolation, and S is set to 3. Note that nearest-neighbor interpolation can be used when computational efficiency is required. In addition, bicubic interpolation can be used for partial replacement when it is necessary to balance computational efficiency and accuracy. The proposed method is compatible with these interpolation methods. We do not consider exploring these tricks here, and scaling the image using bilinear interpolation can achieve remarkable effectiveness. Let $\mathbb{I}$ be the bilinear interpolation operation, and three images are defined as

$$\begin{aligned} x_1 &= \mathbb{I}_{\tau=0.5}(x) \\ x_2 &= \mathbb{I}_{\tau=1}(x) \\ x_3 &= \mathbb{I}_{\tau=2}(x) \end{aligned} \tag{1}$$

where τ is the scaling factor, $\tau = 0.5$ denotes scale down the image by a factor of 2, $\tau = 2.0$ denotes scale up the image by a factor of 2, and $\tau = 1$ means no operation. We apply three parallel branches to extract features at different scales. The branches do not share parameters owing to the training network in a collaborative way. The discrepant branches are in favor of decoupled features. Each branch $J = E \circ G$ could be divided into an encoder E and a decoder G. Each x_s is passed through the encoder and produces semantic features g_s. Then, g_s is passed through the decoder and produces h_s

$$g_s \triangleq E(x_s), \quad h_s \triangleq G(g_s) \tag{2}$$

where h_s is the feature map before the classification layer.

We adopt Resnet34 [50] as the encoder and utilize a modified U-Net [32] decoder. The decoder of U-Net combines high-level semantic features with low-level shallow features by skip connection, which is effective in recovering fine-grained details of objects. The skip connection operation is helpful for remote sensing semantic segmentation. Because pixels of the

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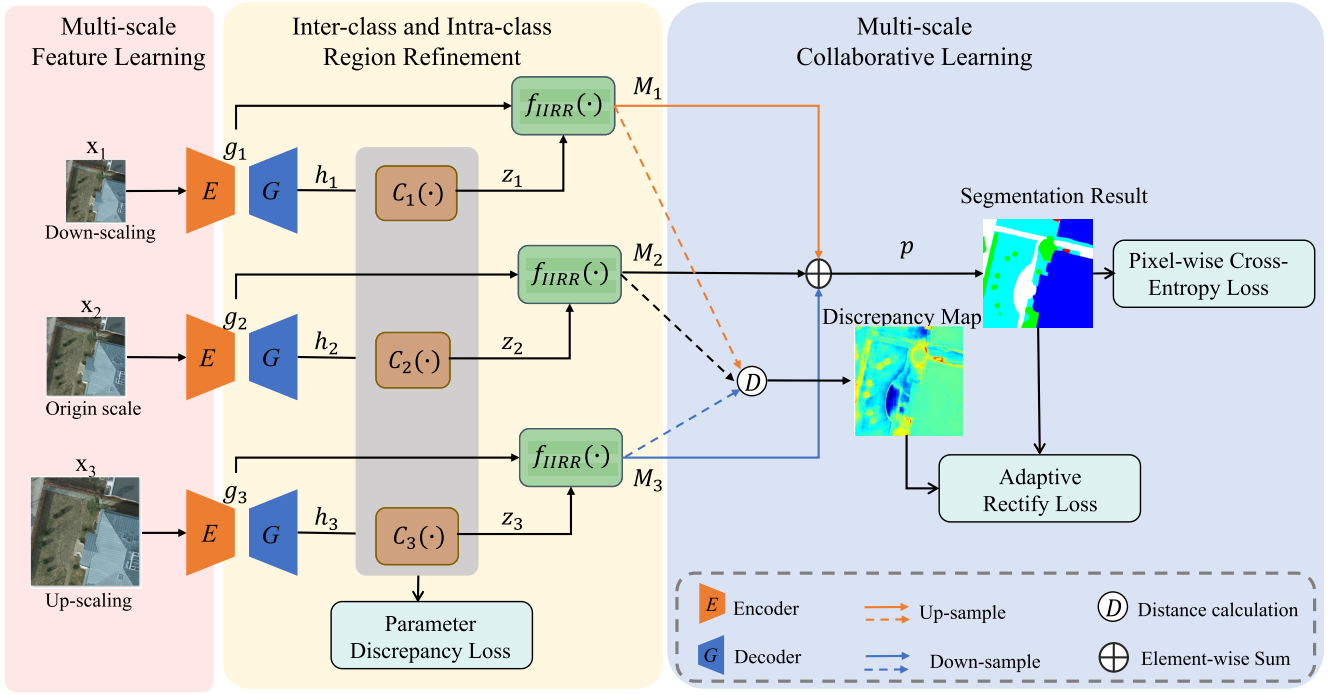


Fig. 2. Overall framework of our proposed method. First, three different scales of remote sensing data are sent to the network simultaneously, which enables each branch to learn MS features. The second part seeks the optimal IIRR $f_{\text{IIRR}}(\cdot)$ to refine scale-aware features. C_1, C_2, and C_3 represent diverse classifiers. The third part is proposed to achieve compatible fusion.

same land objects have similar textural characteristics, fine-grained features are necessary to identify land objects.

B. Inter-Class and Intra-Class Region Refinement

The structure with parallel branches ensures a comprehensive understanding of the input data. However, the redundancy between features in the parallel branch might be high. Attention mechanism can be used to refine features [51], but has a high cost on computation and memory, especially for large size feature maps. Some methods [52], [53] set attention modules after encoder modules, and then upsample to the original size and output segmentation results. Other methods [54], [55] use a gated operator to control the information flow for semantic segmentation. However, this method ignores the propagation through skip operation and may not work well in semantic segmentation for land objects.

In this module, our requirements are threefold. First, enough fine-grained details information to help for land objects recognition, while not containing redundant features too much. Second, the refinement mechanism should be time-efficient. Third, to fuse without feature redundancy, each branch should have discriminative scale-specific features. For this purpose, we propose IIRR, which utilizes the region-wise semantic features to guide the pixel-wise classification feature. We also present the results of the dual attention (DA) [53] module and gated fully fusion (Gated) [55] module comparison with our IIRR module.

For the region-wise semantic features, we utilize high-level semantic features from the last level of the encoder. A value of the high-level semantic features represents the 32 × 2 region in the original image. The region-wise representations have fewer errors in the process of modeling correlation, since the feature representations of pixels with the same category are very different in complex scenes, as illustrated in Fig. 3(a).

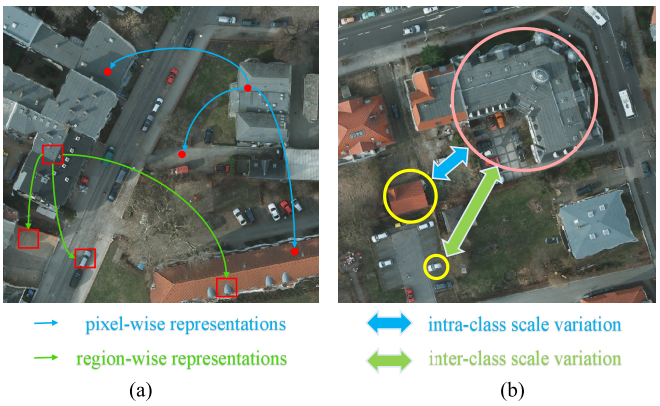


Fig. 3. Illustration of (a) pixel-wise and region-wise representations and (b) inter-class and intra-class scale variation.

Then, we further exploit correlations of region-wise representations to seek the optimal weight M_s for MS features refinement. Land objects in both intra-class and inter-class have different scales. As illustrated in Fig. 3(b), for intra-class, small objects can be detected at scaled-up (2.0×) images while big objects suffer from inconsistency. For inter-class, some classes, which have more texture features, need to be detected at high resolution, while other classes need more context information for accurate predictions. However, the

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feature extractor can only perceive a specific range, and land objects that are too large or too small make the network generate ambiguous features. Considering the intra-class and inter-class scale variation, we model the intra-class refinement and inter-class refinement as DA modules [53] to emphasize compatible ones and reduce the ambiguous features.

Given the high-level semantic feature map g_s, we produce a class activation map $\eta_s \in \mathbb{R}^{C\times H\times W}$ via a convolution layer and a normalization layer, where C, H, and W are the numbers of classes, height, and width of the feature map, respectively. We use three 1×1 convolutions to generate three feature maps ϕ_s, θ_s, and γ_s. Then, we calculate M_s as

$$\begin{aligned} M_s &= \left(M_s^{\text{inter}} + M_s^{\text{intra}}\right) \odot z_s \\ M_s^{\text{intra}} &= f_R(\text{softmax}(\theta_s \cdot \phi_s) \cdot \gamma_s) + \eta_s \\ M_s^{\text{inter}} &= f_R\left(\text{softmax}\left(\eta_s \cdot \eta_s^T\right) \cdot f_R(\eta_s)\right) + \eta_s \end{aligned} \tag{3}$$

where f_R is the reshape operation, $\odot$ indicated element-wise multiplication, and z_s is the dense logits feature of a branch, which gains fine-grained details by skip connection. M_s is calculated with region-wise semantic features and avoids pixel-wise computation, which is time-efficient. Meanwhile, the fusion of M_s^{intra} and M_s^{inter} further refines MS features to be more discriminative. IIRR operations reduce feature redundancy in a more efficient and robust manner.

C. MS Collaborative Learning

Based on the proposed IIRR, compatible ones can be persisted, and less useful ones can be suppressed. However, lack of diversity causes some areas whose results are worse at all scales. To enforce the branches to be diverse for distinct scale views, we propose an MCL strategy. The strategy contains two collaborative losses to obtain the heterogeneous features from the different branches as well as gain accurate results after diversity constraints. In this process, the diversity relationship is adjusted adaptively to rectify the feature redundancy area. The feature compatibility area is preserved for accurate results.

The classifiers in each scale are enforced to have possibly diverse parameters to provide us with distinct views according to the co-training strategy. The distinct views help the module obtain heterogeneous features from different branches. We enforce the parameters of classifiers to be diverse by minimizing their similarity. We use the parameter discrepancy loss to enforce the classifiers have diverse parameters. Set $\overrightarrow{\alpha}_s$ and $\overrightarrow{\alpha}_{s'}$ as the parameters of the convolution filters of C_s and $C_{s'}$, where $s, s' \in \{1, \ldots, S\}$ and $s' \neq s$; we calculate the cosine distances between them, and the parameter discrepancy loss is

$$L_{\text{pd}} = \frac{1}{S}\sum_{s=1}^{S}\sum_{s',s'\neq s}^{S} \frac{\overrightarrow{\alpha}_s \cdot \overrightarrow{\alpha}_{s'}}{\|\overrightarrow{\alpha}_s\| \cdot \|\overrightarrow{\alpha}_{s'}\|}. \tag{4}$$

For the refined features of each branch, we feed them into the classifier C_s to obtain the categorical score map $z_s = C_s(h_s)$. The final segmentation result p is obtained by summing up all predictions p_s

$$p = \frac{1}{S}\sum_{s=1}^{S} p_s, \quad p_s = \mathbb{I}(M_s). \tag{5}$$

Note that the sum operation here is a feasible and straightforward option, since feature diversity is constrained by back-propagation of the losses. The parameter discrepancy loss constrains the diversity of MS feature representations, and the adaptive rectify loss corrects the uncertainty segmentation results at the pixel level. The losses replace designing additional modules to enhance diversity. Therefore, in forwarding propagation, the sum operation of features is a feasible and straightforward choice.

Besides, we compute the discrepancy D to adaptively calibrate the diversity relationship by contrasting the predictions. For prediction p_s and $p_{s'}$, we calculate the distance metric $D(p_s, p_{s'})$ to measure the pixel-wise discrepancy between the predictions. We adopt cosine distance between predictions to compute the pixel-wise discrepancy at position i by

$$D\left(p_s^i, p_{s'}^i\right) = 1 - \left\langle \frac{p_s^i}{\|p_s^i\|}, \frac{p_{s'}^i}{\|p_{s'}^i\|} \right\rangle \tag{6}$$

with D takes the form as

$$D = \frac{1}{S}\sum_{s}^{S}\sum_{s',s'\neq s}^{S} D(p_s, p_{s'}). \tag{7}$$

Then, we generate the offset O of the segmentation result, which is calculated by Kullback–Leibler (KL) divergence between prediction distribution P and truth distribution P'. O can be formulated as follows:

$$\begin{aligned} O &= D_{KL}\left(P'(x)||P(x)\right) \\ &= P'(x)\log P'(x) - \mathbb{E}_{x\sim P'}\left[\log P(x)\right]. \end{aligned} \tag{8}$$

Finally, an element-wise multiplication (indicated by the symbol $\odot$) is performed between the offset O and the discrepancy D. As a result, the adaptive rectify loss L_{ar} can be calculated by

$$L_{\text{ar}} = \sum O \odot D. \tag{9}$$

The first term of O is a constant, so we can simplify the loss to

$$L_{\text{ar}} = \sum_{x}\left(-\mathbb{E}_{x\sim P'}\left[\log P(x)\right]\right) \odot D. \tag{10}$$

The adaptive rectify loss L_{ar} is used for adaptively rectify of uncertain areas. Specifically, D reflects the discrepancy in the MS predictions. If the prediction results from different perspectives are very different, it proves the uncertainty of the predictions to a certain extent. We extend the uncertainty estimate with an adaptive weight O. On the one hand, if the offset O is small, indicating that MS predictions are compatible, it encourages the model to ignore the discrepancy coming from different scales. On the other hand, if the offset O is large, indicating the features of the same object do not have similar joint distribution between MS branches, it should be largely weighted and encourage the model to focus on this area. Therefore, each pixel is differently weighted to adaptively rectify the segmentation result.

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Algorithm 1 Training Algorithm of MANet

Input: Images and corresponding labels $\{x(i), y(i)\}_{i=1}^{N}$, N is the number of training images, learning rate β, inter- and intra-class region refinement f_{IIRR}, sesmantic segmentaion function Θ (feature extraction function Θ_F, classifiers function Θ_C).

Load parameters from initial parameters.
while Segmentaion function Θ not converged **do**
  Calculate multi-scale images x_s according to Eq.1
  Feedforward x_s into endocer and decoder to obtain features g_s and h_s according to Eq.2;
  Generate score map z_s by classifiers.
  Aggregate the reginement map $M_s = f_{IIRR}(g_s, z_s)$;
  Predict segmentation result p using Eq.5;
  Aggregate discrepancy D of multi-scale predictions and the offset O of the segmentation result;
  Compute pixel-wise cross-entropy loss L_{ce}, adaptive rectify loss L_{ar}, parameter discrepancy loss L_{pd} and the final segmentation loss L;
  Compute gradients and update parameters:

$$\Theta = \Theta - \beta \frac{\partial L}{\partial \Theta}$$

  specifically:

$$\Theta_C = \Theta_C - \beta \frac{\partial (L_{ce} + L_{pd} + L_{ar})}{\partial \Theta_C}$$

$$\Theta_F = \Theta_F - \beta \frac{\partial (L_{ce} + L_{ar})}{\partial \Theta_F}$$

end while

Output: Optimized segmentation function Θ.

D. Optimization Objective

Overall, we use three loss functions to train MANet, i.e., the pixel-wise cross-entropy loss L_{ce}, the parameter discrepancy loss L_{pd}, and the adaptive rectify loss L_{ar}. Specifically, given an input image x, the segmentation result p, and the label map y, the pixel-wise cross-entropy loss can be calculated by

$$L_{\text{ce}} = -\frac{1}{|x|} \sum_{i \in x} y^i \log\left(\text{softmax}\left(p^i\right)\right) \tag{11}$$

where i is the pixel in the image.

The network weights, as well as the discrepancy classifier, can be optimized with a joint loss function L in an end-to-end style. With λ_1 and λ_2 are utilized to balance the optimization process, the full optimization objective is

$$L = L_{\text{ce}} + \lambda_1 L_{\text{pd}} + \lambda_2 L_{\text{ar}}. \tag{12}$$

This optimization can effectively improve MS predictions. The parameter discrepancy loss L_{pd} ensures the classifiers capture possibly different features for each branch. Meanwhile, the pixel-wise cross-entropy loss L_{ce} requires each branch to make the same prediction of any scale features. Consequently, the two losses actually enforce that the whole model needs to capture the basic aspects of features at all scales, just as humans can identify objects at any scale. Finally, L_{ar} is used to adjust improper areas and preserve compatibility areas for accurate results. Therefore, the proposed MANet can effectively combine MS features and improve semantic segmentation.

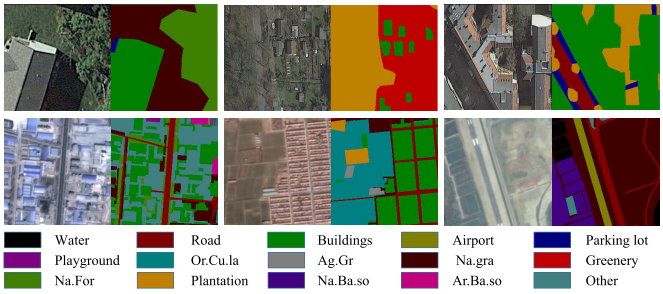


Fig. 4. Examples of pairs of images and corresponding ground truth from the NAIC dataset. The shorthands mean: ordinary cultivated land (Or.Cu.la), agricultural greenhouse (Ag.Gr), natural grassland (Na.gra), natural forest (Na.For), natural bare soil (Na.Ba.so), artificial bare soil (Ar.Ba.so.), and other.

IV. Experimental Results and Analysis

In this section, we first introduce datasets and the experimental setup about implement details. Then, we carry out tests of each module through ablation experiments on Potsdam and NAIC datasets. Finally, we provide comparisons with other MS methods and conclude our experimental validation on three MS variation datasets.

A. Dataset Description

1) Potsdam Dataset: This is a widely used dataset provided by *the International Journal of Photogrammetry and Remote Sensing (ISPRS)* for the 2-D semantic labeling contest.[1] The whole dataset includes 38 true orthophoto (TOP) tiles together and the corresponding normalized digital surface models (nDSMs) data. We utilize 24 tiles for training, and the remaining 14 are used for testing as setting in [14], [56], and [57]. Besides, each training tile is cropped randomly to 256 patches with a size of 512 × 512 pixels in an overlapping manner.

2) NAIC Dataset: The NAIC dataset is a multiresolution dataset proposed by the National Artificial Intelligence Contest[2] in China, collected by multiple satellites, labeled and constructed by the Peng Cheng Laboratory and other co-organizers. The ground sampling distance (GSD) is 0.1–4 m. The size of each image is 256 × 256 pixels. The ground-truth maps are labeled into 15 lands used categories: water, road, buildings, airport, parking lot, playground, ordinary cultivated land, agricultural greenhouse, natural grassland, greenery, natural forest, plantation, natural bare soil, artificial bare soil, and other. We will use $C1$–$C15$ to represent the 15 categories. When training, we split the 60 000 images for training and 40 000 images for evaluation. Fig. 4 shows some examples of training image and the corresponding ground truth.

[1] https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-sem-label-potsdam/

[2] http://www.china-ai.ac.cn/

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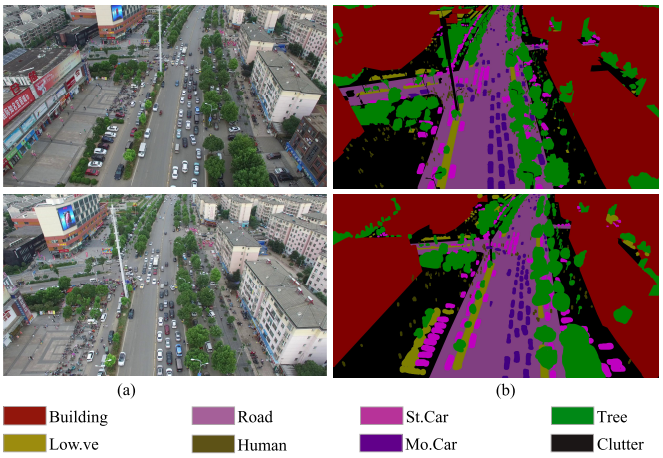


Fig. 5. Example from the UAVid dataset. (a) Training images. (b) Ground truth.

3) UAVid Dataset: This is a high-resolution UAV dataset [15] for semantic segmentation. The top view and side view of each category can be observed in the image. This view brings the challenges of large-scale variation as illustrated in [15]. The dataset contains 30 video sequences. Following the dataset split,[3] we use 15 sequences for training and 5 sequences for validation. The other ten sequences are used for testing. We crop each image to patches with the size of 512 × 512 in an overlapping manner for training. Examples of the training image and its label are presented in Fig. 5.

B. Implement Details

1) Data Augmentation: To avoid overfitting, especially for the ISPRS Potsdam dataset, which has a small number of training samples [58], data augmentation and transfer learning are adopted. Flip and rotation are performed randomly while training each batch. The red green blue infrared (RGBIR) and nDSM of the Potsdam dataset are concatenated into five channels.

2) Hyperparameters: For all datasets, we adopt SGD with momentum set to 0.9 as the optimizer. The initial learning rate is set to 0.0005. We utilize the cosine learning decay policy with a minimum learning rate 5e-8. λ_1 and λ_2 are set to 0.01 and 0.5. The number of training epochs is 20 for the Potsdam dataset and set to 50 for the others. The batch size is set to 10 per GPU for image datasets and set to 1 for the video dataset.

3) Network: The proposed MANet is implemented with PyTorch. To obtain more accurate predictions, we follow effective work[4] and improve it with ASPP [34] and Lovasz Loss [59] as our basic work. ASPP is applied on the top of U-Net decoder, and we set the dilated convolution rate to (1, 6, 12, 18). ResNet34 pre-trained on ImageNet [60] is our backbone for all datasets during entire experiments. The channels of the first convolutional layer are changed according to the input data. We use the same backbone as our proposed model for DeepLabv3+ and PSPNet.[5] For the UAVid video dataset, we send all frames of an image patch to the network per batch. The convolutions and batch normalization layers in IIRR are replaced with 3-D operations to utilize time information.

[3]https://uavid.nl/
[4]https://github.com/selimsef/dsb2018_topcoders
[5]https://github.com/yassouali/pytorch_segmentation

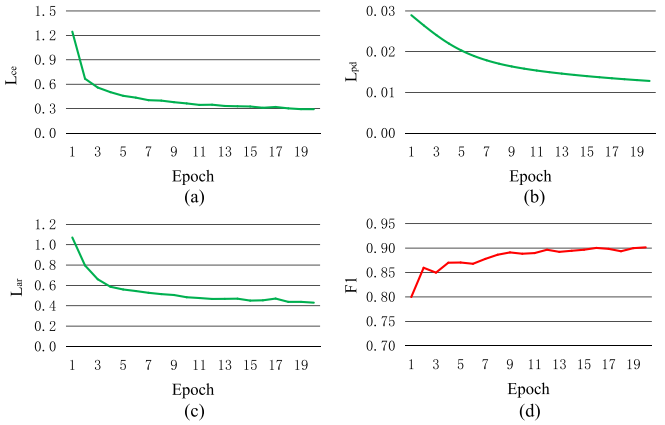


Fig. 6. Training curves (Potsdam dataset). (a) Curve of cross-entropy loss. (b) Curve of parameter discrepancy loss. (c) Curve of adaptive rectify loss. (d) Curve of validation $F1$.

C. Evaluate Metrics

To assess quantitative performance, we use three evaluation metrics that are used in this article: overall accuracy (OA) [2], $F1$ score [61], and mean intersection over union (IoU). tp, tn, fp, and fn represent for the true positives, true negatives, false positives, and false negatives, respectively. Precision P and recall R can be calculated as

$$P = \frac{\text{tp}}{\text{tp} + \text{fp}}, \quad R = \frac{\text{tp}}{\text{tp} + \text{fn}}. \tag{13}$$

Then, OA can be donated as

$$\text{OA} = \frac{\text{tp}}{\text{tp} + \text{fp} + \text{fn} + \text{tn}}. \tag{14}$$

IoU can be defined as

$$\text{IoU} = \frac{\text{tp}}{\text{tp} = \text{fp} = \text{fn}}. \tag{15}$$

$F1$ score is defined as follows:

$$\text{F1} = 2 \times \left(\frac{P \cdot R}{P + R} \right). \tag{16}$$

We calculate the abovementioned metrics to evaluate the segmentation result on three datasets.

D. Training Stability Analysis

To quantitatively evaluate the convergence of joint training with proposed losses and the effectiveness of the proposed method, a further experiment is conducted on the Potsdam dataset. Fig. 6 shows the behavior of each loss and the $F1$ score versus epochs. It can be seen from Fig. 6(a)–(c) that the curves L_{ce}, L_{pd}, and L_{en} decrease rapidly during early training and tend to stabilize after several periods. All loss curves show a downward trend, indicating that the joint training of three losses can steadily converge. We can see from Fig. 6(d) that

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TABLE I
RESULTS ($F1$) OF THE ABLATION RESULTS ON TWO DATASETS

Model	IIRR	MCL	POTSDAM	NAIC
baseline			88.9	74.3
baseline+IIRR	✓		89.5	75.3
baseline+MCL		✓	89.4	75.3
baseline+IIRR+MCL	✓	✓	90.2	75.9

TABLE II
COMPARISON OF IIRR WITH OTHER MODULES ON THE POTSDAM DATASET

Model	OA	F1
MANet(with **IIRR**)	**92.1**	**90.2**
MANet*(with DA module)	90.8	88.2
MANet**(with Gated module)	89.7	85.8

the validation $F1$ gradually increases and is finally stable. This indicates that our proposed method is practicable and adaptive to MS learning.

E. Ablation Study: Effect of IIRR

To verify the effectiveness of the proposed modules, we take U-Net-MS as our baseline, which uses a multinet architecture with average-pooling fusion. We present comparisons of the segmentation results obtained by each module. Table I shows that IIRR improves $F1$ from 88.9% to 89.5% (0.6% ↑) on the Potsdam dataset and from 74.3% to 75.3% (1.0% ↑) on the NAIC dataset, which shows its capability of refining the features to be more discriminative and improve the performance for MS datasets. The improvement in the NAIC dataset is obvious, because the large-scale variation problem in the NAIC dataset is more serious. Table II compares the IIRR module with DA module [53] and Gated module [55], which shows the diversity of the IIRR module for remote sensing semantic segmentation. More qualitative results will be explained in detail.

1) Visualization of IIRR: To further understand the performance of IIRR, we visualize the refinement maps taken by IIRR in Fig. 7. The bright region represents the high response. Different scales have unique concerns, just as the process of human perception. As shown in Fig. 7(b), the weight maps from scale 1 (scale down the original image) have a high response on large-scale buildings, because it is easy to obtain the global context. Fig. 7(c) shows the weight map from scale 2 (original image). We can observe module concentrates on edges or middle scale buildings. In Fig. 7(d), the weight map from scale 3 (scale up the original image) usually focuses on small-scale objects or textured land. Different scale branches perceive different objects on the image both in inter-class and intra-class. Therefore, using these weight maps to refine MS features will make the feature representations become more discriminative.

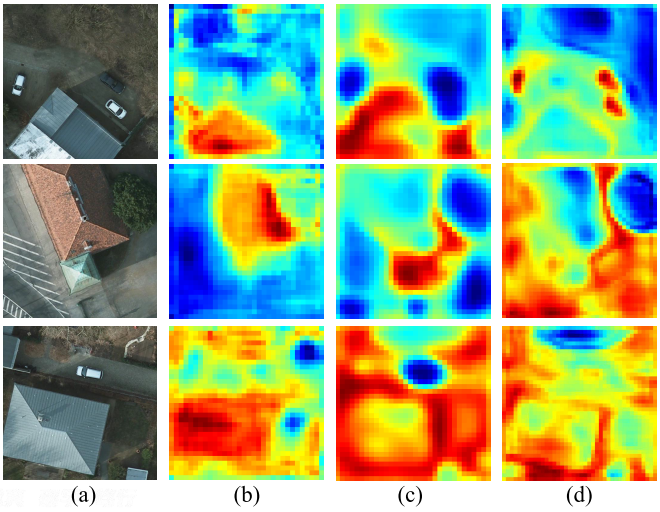


Fig. 7. Partial visualization of feature maps from IIRR. (a) Red green blue (RGB) images. (b) Feature maps from scale 1. (c) Feature maps from scale 2. (d) Feature maps from scale 3.

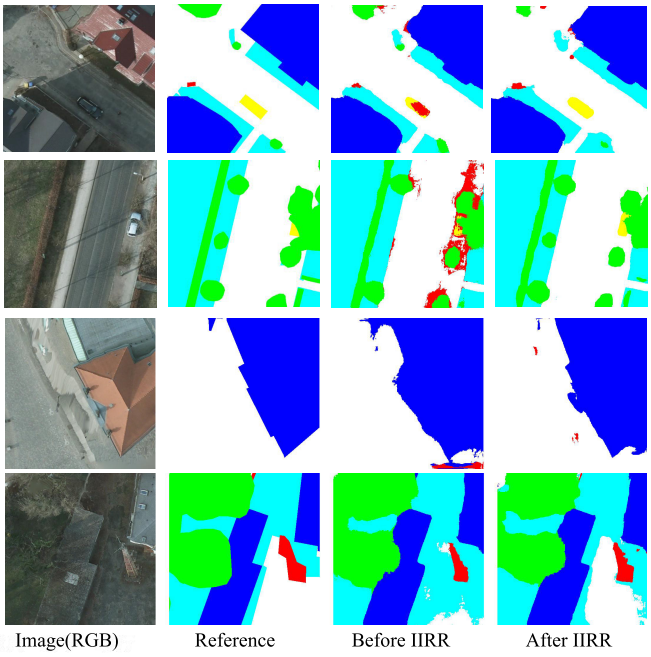


Fig. 8. Comparison of segmentation results before and after using IIRR on the Potsdam dataset.

2) Ablation Experiment on Potsdam Dataset: Fig. 8 shows the performance of applying the IIRR on the Potsdam dataset. The object scales in the high-resolution Potsdam dataset are varied, which leads to confusion of category on the results. However, after utilizing the proposed IIRR, feature redundancy is reduced, and MS features are more discriminative. From the comparisons, we can observe improvements from two aspects. For inter-class fusion noise, as the first and second rows in Fig. 8, IIRR can refine the features to be more discriminative and improve the performance. For intra-class fusion noise, as the last two rows in Fig. 8, IIRR can persist compatible areas and suppress less useful ones during refinement. One can

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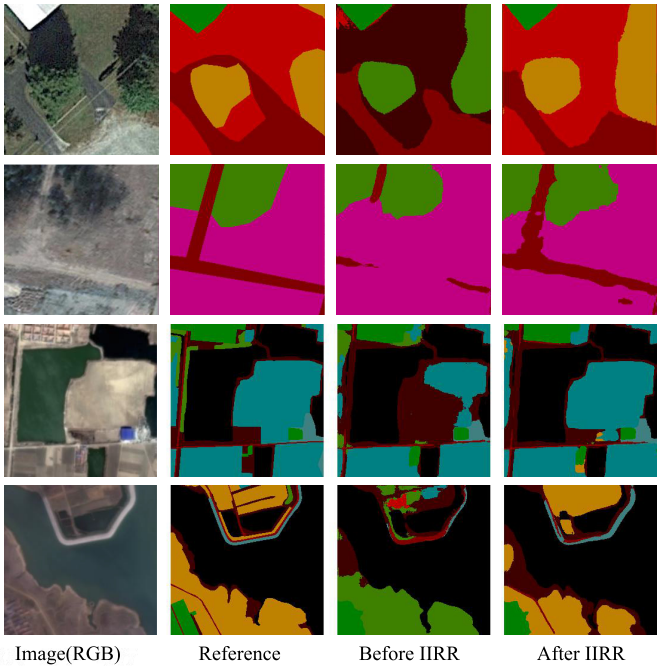


Fig. 9. Comparison of segmentation results before and after using IIRR on the NAIC dataset. In the third column, feature redundancy makes some categories confused and causes ambiguous results. After applying IIRR, the problem is alleviated.

observe that the boundaries of different classes can be explicitly divided. Moreover, the effect of the building, tree, and clutter/background classes can be segmented more accurately. The IIRR module can integrate the information of different scales more adaptively.

3) Ablation Experiment on NAIC Dataset: Tests of applying the IIRR on the multiresolution NAIC dataset are shown in Fig. 9. The confusion of object classes is more serious before applying the IIRR. For inter-class, the obvious observation is the first and last rows in Fig. 9, and $C12$ is classified to $C11$. For intra-class, as the second row of Fig. 9, one can observe that road class is disturbed by the features of other classes, which shows inconsistent prediction of the same class. The same situation also appears in the third row of Fig. 9. We can observe improvements after IIRR is used to help with both inter-class and intra-class. This change is especially obvious in multiresolution datasets.

F. Ablation Study: Effect of MCL

As represented in the third rows of Table I, adding MCL independently to the baseline improves $F1$ from 88.9% to 89.4% (0.5% ↑) on the Potsdam dataset and from 74.3% to 75.0% (0.7% ↑) on the NAIC dataset. It indicates that the performance improvement of MCL is limited. But when applying both IIRR and MCL, the $F1$ score increased by 1.3% and 1.6% on the Potsdam dataset and the NAIC dataset, respectively, which further illustrates the effectiveness of our method. Table III shows the effectiveness of parameter discrepancy loss and adaptive rectify loss for remote sensing

TABLE III

EFFECT OF PARAMETER DISCREPANCY LOSS (L_{pd}) AND ADAPTIVE RECTIFY LOSS (L_{ar}) ON THE POTSDAM DATASET

Model	L_{pd}	L_{ar}	OA	F1
MANet	✓	✓	92.1	90.2
-removing L_{ar}	✓		91.6	89.8
-removing L_{pd}		✓	91.7	89.7

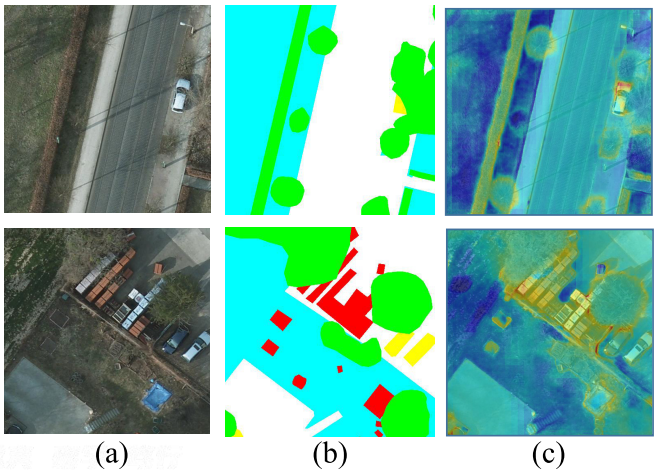


Fig. 10. Partial visualization of discrepancy maps. For each row, from left to right are (a) original images, (b) ground truth, and (c) discrepancy maps.

semantic segmentation. This further demonstrates the collaboration of the losses.

1) Visualization of Discrepancy Map: To further understand how MCL influences the results, we visualize the discrepancy maps on the Potsdam dataset in Fig. 10. We superimpose the difference map with the original image, which can help to understand. The brighter area means the map value is larger than other areas. We can observe that places with large values are mostly regions of edges and large-scale objects, which need global context. This shows the diversity and complementarity of the feature maps from three scale branches.

2) Ablation Experiment on Potsdam Dataset: To visually confirm the effectiveness of the proposed modules, we compare the segmentation results before and after the use of IIRR and MCL on the Potsdam dataset in Fig. 11. The results before IIRR are the outputs of U-Net-MS, which uses a multinet architecture with average-pooling fusion. Although U-Net-MS combines multiple scales to improve the results of U-Net, it suffers from the problem of combining the best predictions with poorer predictions, which leads to noises. The results after IIRR are less noisy than before, because IIRR refines the features to be more discriminative. Nevertheless, for some scenes with complex textures, improving the diversity representation of features is helpful for the recognition of complex scenes. For example, in the first row of Fig. 11, the shadows of tree branches make it difficult to distinguish flat building roofs from the ground. Therefore, MCL with feature diversity constraints improves the noise, making the results

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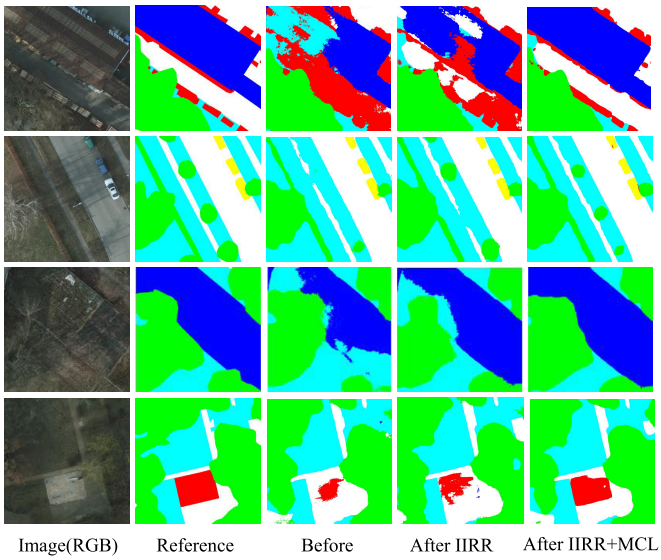


Fig. 11. Comparison of segmentation results before and after using IIRR and IIRR + MCL on the Potsdam dataset.

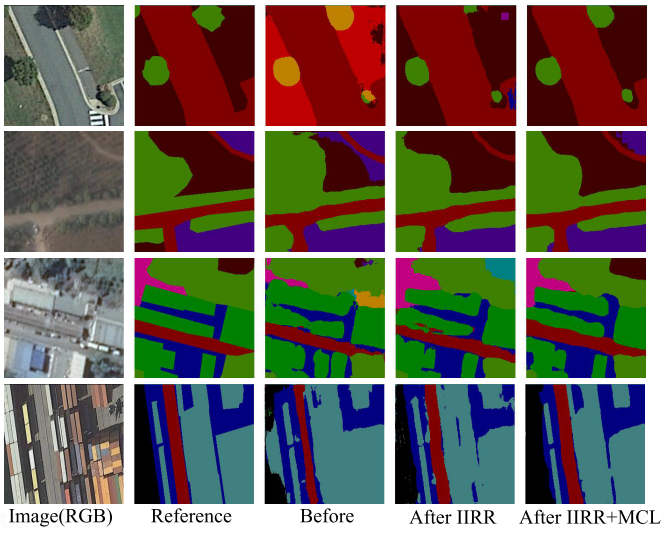


Fig. 12. Comparison of segmentation results before and after using IIRR and IIRR + MCL on the NAIC dataset.

after IIRR + MCL less noisy. MCL enforces the features to be diverse and serves as a contributing factor for MS fusion. Meanwhile, MCL encourages the model to focus on the worse area and improve this part. One can observe that the results can be correct after using MCL to some extent.

3) Ablation Experiment on NAIC Dataset: Fig. 12 shows the effect of applying the IIRR and MCL on the NAIC dataset. IIRR is expected to produce better results than U-Net-MS due to its refinement of scale-aware features. However, the diversity of MS features is limited. After applying both IIRR and MCL, the network can combine discriminative and diversified MS features to enhance the diversity of MS feature representations and improve semantic segmentation. As shown in Fig. 12, in our predictions, the noise area is corrected, and object boundaries are more smooth for many classes. With the introduction of the MCL, the segmentation results are better after using MCL.

TABLE IV

RESULTS IN TERMS OF PER-CLASS $F1$ SCORE, OA, AND AVERAGE $F1$ SCORE (POTSDAM DATASET)

Model	Im.Sur.	Build.	Low.ve.	Tree	Car	F1	OA
DeepLabv3+	88.7	95.0	82.0	81.7	83.8	88.1	86.2
PSPNet	89.5	95.0	82.8	82.6	85.6	88.8	87.1
U-Net	88.8	93.8	83.2	83.4	88.4	89.0	87.5
U-Net-MS	90.0	95.1	84.5	85.7	89.0	91.5	88.9
HRNet-OCR	90.7	95.2	84.5	85.8	90.6	90.3	89.3
HRNet	**91.5**	95.7	84.8	85.9	90.3	90.5	89.6
EfficientPS	89.2	92.9	83.5	84.4	87.1	87.4	90.0
MANet	91.2	**96.0**	**85.7**	**86.6**	**91.3**	**92.1**	**90.2**

G. Qualitative Analysis of the Semantic Segmentation Results

We evaluate the performance of our MANet on the test set. Also, the size of each test image is the same as the train set. For image datasets, we illustrate the effectiveness of this approach by comparing with other MS models, including U-Net [32], PSPNet [31], DeepLabV3+ [5], high-resolution network (HRNet) [62], HRNet-object-contextual representations (OCR) [63], and our baseline (U-Net-MS). For the video dataset, we compare the proposed model with other video models, including fully convolutional network with 8 strides (FCN-8s) [64], U-Net [32], MSD [15], and U-Net-MS. All these models are implemented with the ResNet34 as the backbone.

1) Results on Potsdam Dataset: Table IV presents the performance on the Potsdam dataset. From Table IV, we can observe that our proposed MANet shows an advantage and outperforms the U-Net-MS 1.3% on the average $F1$ score and 0.6% on the OA. OCRNet [63] exploits the context of objects to extract MS information. HRNet-OCR feeds the output of HRNet [62] into the OCR module, which constructs an efficient MS fusion method. We compare the proposed method with HRNet-OCR, and experiments show that the $F1$ score is improved by 1.8%, and the OA is improved by 0.9%. This is because HRNet-OCR enhances object contextual feature representation, while our method improves contextual feature representation as well as enhances discriminability and diversity of feature representation. EfficientPS [65] proposes two-way feature pyramid networks (FPNs), which can encode and aggregate MS features bidirectionally. Also, the proposed semantic segmentation head can fuse long-range context. We compare the proposed method with EfficientPS under the same task setting, and experiments show that the $F1$ score is improved by 4.7%, and the OA is improved by 0.2%. Our method is more advantageous in the $F1$ score. This is because the $F1$ score considers not only the precision but also the recall. Our method constrains the discriminability and diversity of features, which enriches the feature space and improves recall.

Examples of testing results are shown in Fig. 13. Note that in Fig. 13(d), the output of U-Net-MS is noisy, because combining the best prediction with the poor prediction results in noise. The proposed method addresses this issue, which employs IIRR and MCL to mitigate fusion noise, thus

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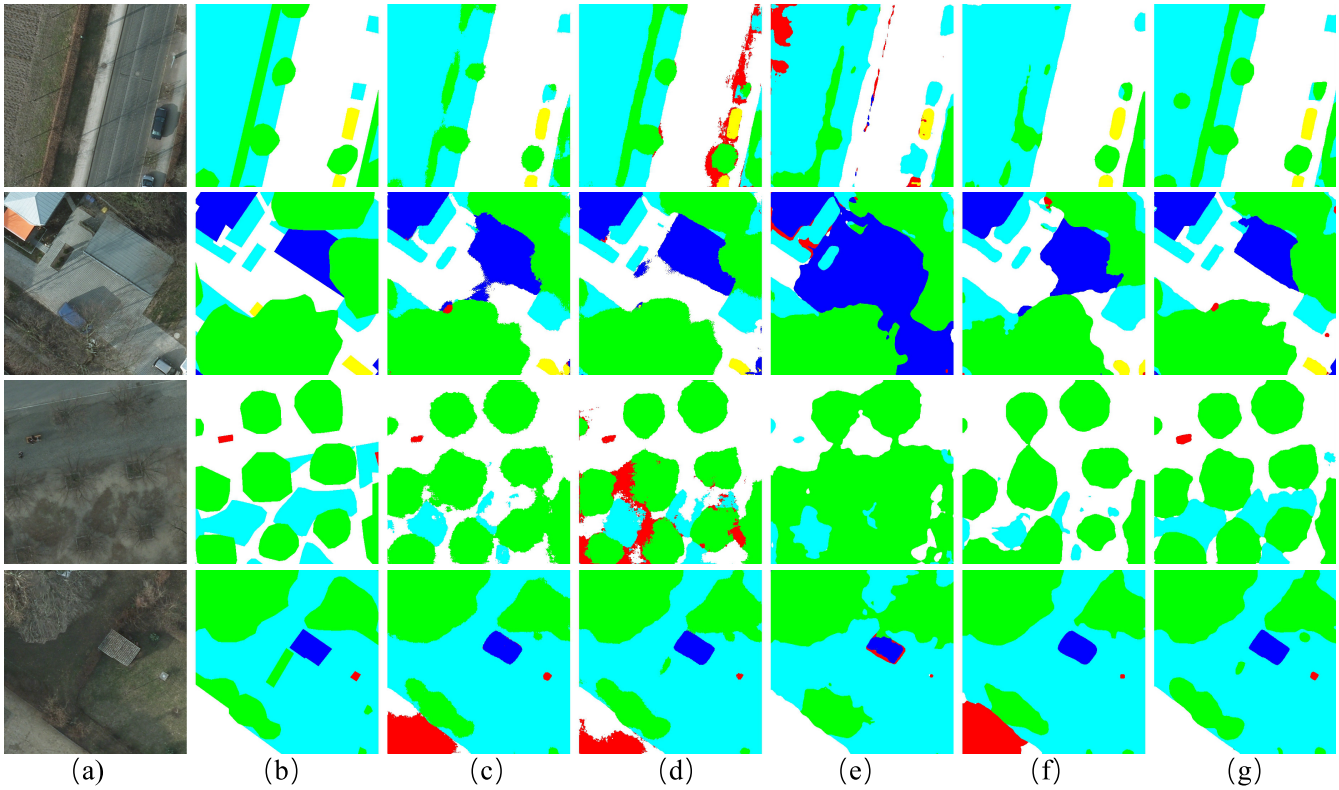


Fig. 13. Segmentation results of the Potsdam test dataset. (a) Images (RGB). (b) Ground truth. (c) U-Net. (d) U-Net-MS. (e) DeepLabV3+. (f) PSPNet. (g) MANet.

TABLE V
RESULTS IN TERMS OF PER-CLASS $F1$ SCORE, OA, AND AVERAGE $F1$ SCORE (NAIC DATASET)

Model	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	OA	F1
PSPNet	80.1	78.3	87.4	0.0	78.7	73.4	89.3	83.1	67.1	69.9	77.4	67.1	74.3	69.6	54.3	78.7	70.0
DeepLabv3+	80.1	79.0	87.6	0.0	79.1	76.0	89.4	83.8	67.1	69.0	77.5	67.8	75.2	71.1	55.5	78.9	70.6
U-Net	83.9	82.0	88.8	0.0	83.1	83.1	90.0	84.7	70.1	74.4	79.6	70.0	78.3	74.0	61.3	81.2	73.5
U-Net-MS	84.7	83.6	89.8	0.2	84.3	84.2	90.2	85.8	70.7	75.0	80.0	70.1	78.7	75.3	62.8	81.9	74.3
HRNet	84.7	83.3	90.3	0.0	83.8	80.4	90.9	86.3	70.2	74.3	80.0	71.5	77.4	74.3	61.3	82.4	73.9
HRNet-OCR	85.0	83.4	**90.3**	0.0	84.3	77.7	**90.9**	**86.6**	**70.8**	74.7	80.2	**71.6**	77.8	74.5	**65.0**	**82.7**	74.2
MANet	**85.1**	**83.4**	90.0	**14.3**	**84.9**	**86.2**	90.1	86.4	70.7	**76.9**	**80.3**	70.9	**79.4**	**76.5**	64.4	82.2	**75.9**

improving performance. Besides, one can observe that the prediction maps of MANet have more accurate segmentation results whatever in inter-class objects or the intra-class objects compared with other approaches. Besides, the proposed model performs better, in detail, than compared models because of the weighted combination across the different scales of predictions, which integrates the information of differently scaled feature maps properly. Moreover, not only the detailed information is captured but also the distinction of categories with different scales is recognized well. Therefore, the proposed MANet outperforms other models.

2) Results on NAIC Dataset: Table V presents the segmentation results on the NAIC dataset. Because there are multiresolution images in the dataset, the large-scale variation images can affect the accuracy of predictions. However, our proposed MANet exploits the correlation among different scales and achieves better results than others. Specifically, our method outperforms the U-Net-MS 1.6% on the average $F1$ score and 0.3% on the OA. Fig. 14 shows the prediction maps generated by each method. Note that in Fig. 13(c), the output of U-Net has some noise because of the lack of MS features in the network. For example, in the first row,

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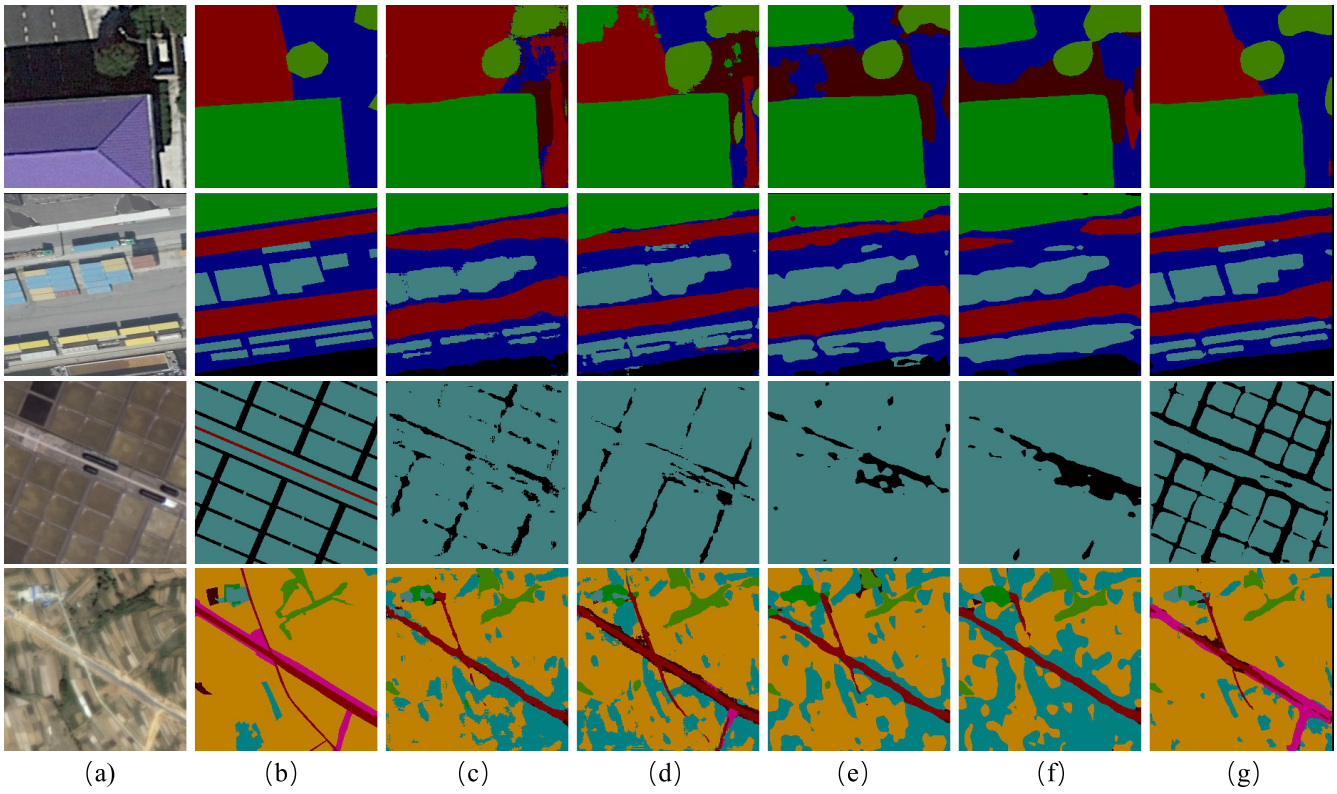


Fig. 14. Segmentation results produced by the compared approaches on the NAIC dataset. (a) Images. (b) Groud truth. (c) U-Net. (d) U-Net-MS. (e) DeepLabV3+. (f) PSPNet. (g) MANet.

TABLE VI
RESULTS IN TERMS OF PER-CLASS IOU AND MEAN IOU (UAVID DATASET)

	Model	Building	Tree	Clutter	Road	Low.Ve	St.Car	Mo.Car	Human	MIOU
UAVid10	FCN-8s [64]	64.3	63.8	33.5	57.6	28.1	8.4	29.1	0.0	35.6
	U-Net [32]	70.7	67.2	36.1	61.9	32.8	11.2	47.5	0.0	40.9
	MSD [15]	**80.9**	**75.5**	46.3	66.7	47.9	22.3	**56.9**	4.2	50.1
	MANet	79.2	72.5	**53.1**	**74.3**	**53.7**	**30.8**	56.6	**28.6**	**56.1**
UAVid2020	U-Net-MS	79.6	69.4	55.1	75.3	52.0	29.8	59.0	16.7	54.6
	MSD [15]	**79.8**	**74.5**	**57.0**	**74.0**	55.9	32.1	62.9	19.7	57.0
	MANet	78.7	73.1	56.9	72.8	**55.9**	**37.3**	**69.0**	**26.7**	**58.8**

complex factors such as shadows interfere with recognizing the scene. This is because U-Net incorporates fine-grained features in the decoding stage. Our method preserves fine-grained features, as well as incorporating high-level semantic features to improve discriminability and diversity. These specially designed priors help the model converge to a better solution and reduce noise in the results. Besides, one can observe that the prediction map of MANet has more accurate segmentation results at different resolutions compared with other approaches. In more detail, the areas contained textured land, and the edges of objects are recognized better than other methods. Meanwhile, predictions of large-scale land objects are also predicted better.

3) Results on UAVid Dataset: We also test MANet on the UAVid dataset taken by an oblique UAV. For this view, the scale of the object changes at different distances. MANet can reduce the impact of large-scale variation. Table VI shows the experimental results of MANet that outperforms all compared methods. We validate the effectiveness on UAVid10, which has 30 sequences and UAVid2020, which adds extra 12 sequences to further strengthen the dataset. From Table VI, we can observe that our proposed model outperforms the MSD 6.0% on the mean IoU of UAVid10 and 1.8% on the mean IoU of UAVid2020. The IoU for the human class is still poor but higher than other methods. The comparisons in segmentation of the MSD, baseline (U-Net-MS), and MANet are shown in Fig. 15. MSD is an MS approach for semantic segmentation tasks. We compare the results of MSD with the proposed method. The scale of objects, such as cars, varies greatly from the bottom to the top of the image. Also, our method

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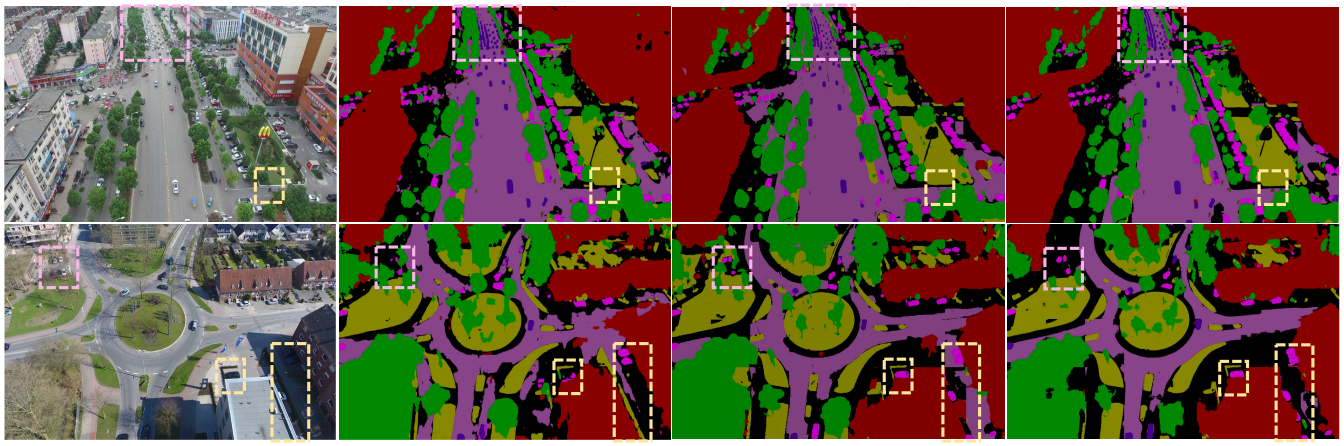


Fig. 15. Segmentation results compared with approaches on the UAVid test dataset. (a) Images. (b) MSD. (c) U-Net-MS. (d) MANet.

is more advantageous for MS recognition. This is because the proposed method considers the discriminability and diversity of MS feature representations. MSD uses three streams to extract MS features and aggregate for prediction, which shows a certain effect on the representation of MS features. Our method further optimizes the discriminability and diversity of MS feature representations to improve predictions.

V. Conclusion

In this article, an MANet is presented to learn relations between different scales by enhancing the discriminative and diverse MS representations for remote sensing data. Considering feature redundancy caused by fusion, IIRR is proposed to make MS features more discriminative. In addition, features of different scales are expected to be diverse. Thus, discrepancy classifiers are designed from diverse MS views. The discrepancy classifiers enhance the diversity of MS feature representations for semantic segmentation. Then, we measure the discrepancy map of the MS predictions and minimize the uncertainty on misclassified regions to adaptively rectify the segmentation results at the pixel level. The proposed approach is evaluated on the Potsdam dataset, the NAIC dataset, and the UAVid video dataset. Experiments show the effectiveness of the proposed approach on both image and video datasets. Future work can combine NAS methods to represent MS features and design network structures.

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References

[1] Z. Huang *et al.*, "CCNet: Criss-cross attention for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Jul. 3, 2020, doi: 10.1109/TPAMI.2020.3007032.

[2] X. Liu *et al.*, "Deep multiview union learning network for multisource image classification," *IEEE Trans. Cybern.*, early access, Nov. 5, 2020, doi: 10.1109/TCYB.2020.3029787.

[3] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[4] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.

[5] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[6] J. Jiang, J. Liu, J. Fu, X. Zhu, Z. Li, and H. Lu, "Global-guided selective context network for scene parsing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1752–1764, Apr. 2022.

[7] J. Wang *et al.*, "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[8] G. Li, L. Li, H. Zhu, X. Liu, and L. Jiao, "Adaptive multiscale deep fusion residual network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8506–8521, Nov. 2019.

[9] X. Liu *et al.*, "Deep multiple instance learning-based spatial–spectral classification for PAN and MS imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 461–473, Jan. 2018.

[10] L. Mou, Y. Hua, and X. X. Zhu, "A relation-augmented fully convolutional network for semantic segmentation in aerial scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12416–12425.

[11] L. Mou and X. X. Zhu, "Vehicle instance segmentation from aerial image and video using a multitask learning residual fully convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 6699–6711, Nov. 2018.

[12] T. Jagdhuber *et al.*, "Physics-based modeling of active and passive microwave covariations over vegetated surfaces," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 788–802, Feb. 2019.

[13] X. Liu, L. Jiao, X. Tang, Q. Sun, and D. Zhang, "Polarimetric convolutional network for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 3040–3054, May 2019.

[14] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.

[15] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, Jul. 2020.

[16] L. Jiao, D. Wang, Y. Bai, P. Chen, and F. Liu, "Deep learning in visual tracking: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 30, 2021, doi: 10.1109/TNNLS.2021.3136907.

[17] T. Lindeberg, *Scale-Space Theory in Computer Vision*, vol. 256. Springer, 2013.

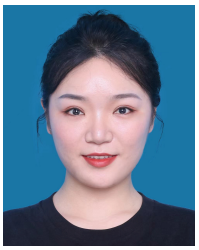
[18] L. Jiao, J. Gao, X. Liu, F. Liu, S. Yang, and B. Hou, "Multi-scale representation learning for image classification: A survey," *IEEE Trans. Artif. Intell.*, early access, Dec. 14, 2021, doi: 10.1109/TAI.2021.3135248.

[19] Y. Jia, C. Huang, and T. Darrell, "Beyond spatial pyramids: Receptive field learning for pooled image features," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3370–3377.

[20] W. Hackbusch, *Multi-Grid Methods and Applications*, vol. 4. Springer, 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:14:50 UTC from IEEE Xplore. Restrictions apply.

[21] D. Terzopoulos, "Image analysis using multigrid relaxation methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-8, no. 2, pp. 129–139, Mar. 1986.
[22] L. Jiao, J. Pan, and Y. Fang, "Multiwavelet neural network and its approximation properties," *IEEE Trans. Neural Netw.*, vol. 12, no. 5, pp. 1060–1066, Sep. 2001.
[23] S. Tan, L. Jiao, and I. A. Kakadiaris, "Wavelet-based Bayesian image estimation: From marginal and bivariate prior models to multivariate prior models," *IEEE Trans. Image Process.*, vol. 17, no. 4, pp. 469–481, Apr. 2008.
[24] Y. Wang, J. Zhang, M. Kan, S. Shan, and X. Chen, "Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12275–12284.
[25] S. Yang and G. Peng, "Attention to refine through multi scales for semantic segmentation," in *Proc. Pacific Rim Conf. Multimedia*. Springer, 2018, pp. 232–241.
[26] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.
[27] A. Tao, K. Sapra, and B. Catanzaro, "Hierarchical multi-scale attention for semantic segmentation," 2020, *arXiv:2005.10821*.
[28] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," *ISPRS J. Photogram. Remote Sens.*, vol. 145, pp. 78–95, Nov. 2018.
[29] H. Chen, Y. Jin, G. Jin, C. Zhu, and E. Chen, "Semisupervised semantic segmentation by improving prediction confidence," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 29, 2021, doi: 10.1109/TNNLS.2021.3066850.
[30] D. Lin *et al.*, "ZigZagNet: Fusing top-down and bottom-up context for object segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7490–7499.
[31] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[32] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Springer, 2015, pp. 234–241.
[33] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[34] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[35] J. Fu, J. Liu, J. Jiang, Y. Li, Y. Bao, and H. Lu, "Scene segmentation with dual relation-aware attention network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2547–2560, Jun. 2021.
[36] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. ACCV*, 2018, pp. 150–165.
[37] H. Ding, X. Jiang, B. Shuai, A. Q. Liu, and G. Wang, "Semantic segmentation with context encoding and multi-path decoding," *IEEE Trans. Image Process.*, vol. 29, pp. 3520–3533, 2020.
[38] Z. Huang, C. Wang, X. Wang, W. Liu, and J. Wang, "Semantic image segmentation by scale-adaptive networks," *IEEE Trans. Image Process.*, vol. 29, pp. 2066–2077, 2020.
[39] X. Zhang *et al.*, "DCNAS: Densely connected neural architecture search for semantic image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13951–13962.
[40] Z.-H. Zhou and M. Li, "Tri-training: Exploiting unlabeled data using three classifiers," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 11, pp. 1529–1541, Nov. 2005.
[41] K. Nigam and R. Ghani, "Analyzing the effectiveness and applicability of co-training," in *Proc. 9th Int. Conf. Inf. Knowl. Manage. (CIKM)*, 2000, pp. 86–93.
[42] C. Yu, J. Wang, C. Gao, G. Yu, C. Shen, and N. Sang, "Context prior for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12416–12425.
[43] M. Liu, L. Jiao, X. Liu, L. Li, F. Liu, and S. Yang, "C-CNN: Contourlet convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2636–2649, Jun. 2021.
[44] S. H. Gao, M. M. Cheng, and K. Zhao, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.
[45] D. Xu, X. Alameda-Pineda, W. Ouyang, E. Ricci, X. Wang, and N. Sebe, "Probabilistic graph attention network with conditional kernels for pixel-wise prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2673–2688, May 2022.
[46] W. Zhao and S. Du, "Learning multiscale and deep representations for classifying remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 113, pp. 155–165, Mar. 2016.
[47] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.
[48] L. Jiao *et al.*, "New generation deep learning for video object detection: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 3, 2021, doi: 10.1109/TNNLS.2021.3053249.
[49] A. Kundu, V. Vineet, and V. Koltun, "Feature space optimization for semantic video segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3168–3175.
[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[51] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.
[52] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[53] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[54] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5229–5238.
[55] X. Li, H. Zhao, L. Han, Y. Tong, and K. Yang, "GFF: Gated fully fusion for semantic segmentation," 2019, *arXiv:1904.01803*.
[56] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.
[57] F. Zhou, R. Hang, and Q. Liu, "Class-guided feature decoupling network for airborne image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2245–2255, Mar. 2021.
[58] X. Chen *et al.*, "Adaptive effective receptive field convolution for semantic segmentation of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3532–3546, Apr. 2021.
[59] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovasz-softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4413–4421.
[60] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.
[61] B. Yu, L. Yang, and F. Chen, "Semantic segmentation for high spatial resolution remote sensing images based on convolution neural network and pyramid pooling module," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 9, pp. 3252–3261, Sep. 2018.
[62] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.
[63] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 173–190.
[64] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[65] R. Mohan and A. Valada, "EfficientPS: Efficient panoptic segmentation," *Int. J. Comput. Vis.*, vol. 129, pp. 1551–1579, Feb. 2021.



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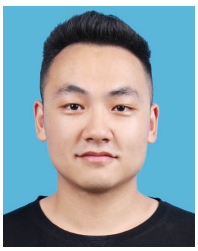
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